Personalized AI for Parkinson Patients

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LASIGE inclusive human-computer interaction

Motivation & Context

- **The Clinical Barrier:** Short consultations force clinicians to rely on patients imperfect recall of symptoms, while smartwatch data cannot be directly interpreted by patients or LLMs.

10 to 15 minutes only! →



- **Research Question**

How does a large language model summarise long-term wearable sensor data across different representations, and can it generate a clinically efficient and safe conversational narrative for Parkinson's disease patients?

Methodology

- **Dataset:** 3 continuous left wrists free-living recordings around 307 hours total.
- **Deterministic DSP:** extracting Signal Vector Magnitude (SVM), tremor, bradykinesia and dyskinesia.
- **Evaluated Representations:**
  - Raw: Unfiltered 3.5s triaxial acceleration(X, Y, Z)
  - SVM: Full-recording scalar dynamic magnitude ($SVM_{\mathrm{dyn}}(t) = \sqrt{x^2 + y^2 + z^2}$)
  - Multi-Symptom Summaries (MSSC): Structured metric summaries of computed symptom percentages
- **Prompt Design:** Enforces a consultation agenda for 10–15 min visits: (1) *Your Week* (overview); (2) *For Reflection* (episodic recall); (3) *Bring to the Doctor* (first-person statement)
- **Evaluation:** Qualitative analysis across 9 runs (3 datasets $\times$ 3 representations) assessing factual grounding, temporal confabulation, and clinical framing

Findings

- **RAW -** Preserves raw signal, but lacks semantic structure; LLM refused quantitative analysis
- **SVM -** Captures broad activity, but discards spectral features, leading to schedule hallucinations.
- **MSSC -** Structured metric summaries enable high verbal correspondence with computed symptoms.



Created by Andre Buand from Noun Project



Created by Hanbai from Noun Project

Discussion

- LLM reliability requires upstream abstraction. SVM strips spectral cues (masking symptoms and fabricating times); structured metrics (MSSC) provide the semantic scaffold required for faithful verbalization.
- Medication questions stemmed strictly from prompt instructions. Interfaces must explicitly signal that reflection cues are conversational prompts, not sensor-verified facts.

- **Core Takeaway:**

Data representation shapes narrative behaviour. Structured summaries improve correspondence with supplied metrics, but prompt design determines which clinical interpretations are invited.

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